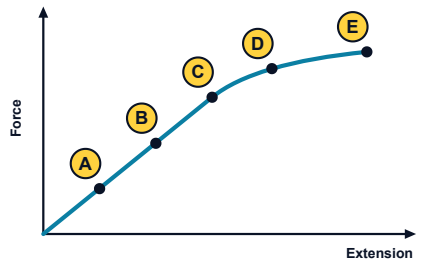
DO NOW, AO1 REVISION

Forces and elasticity

Forces
Paper 2, Combined and Separate
20 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 9 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**Give three ways in which the shape of an object can be changed.**	Stretching, compressing or bending	☐
2 ★	**What is elastic deformation?**	The object returns to its original shape when the forces are removed.	☐
3 ★	**What is inelastic deformation?**	The object does not return to its original shape when the forces are removed.	☐
4 ★	**State how the extension of a spring depends on the force applied, below the limit of proportionality.**	The extension is directly proportional to the force.	☐
5	**Describe the relationship between the force on an elastic object and its extension.**	As the force increases, the extension increases.	☐
6 ★	**What is the limit of proportionality of an elastic object?**	The point beyond which extension is no longer directly proportional to force.	☐
7	**Which letter shows the limit of proportionality on this force–extension graph?** 	C	☐
8	**State the standard unit of force.**	Newton (N)	☐
9 ★	**State the standard unit of spring constant.**	Newton per metre (N/m)	☐
10 ★	**Define spring constant.**	The force needed to stretch (or compress) a spring by 1 m.	☐

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	QUESTION	ANSWER	✓
11	**State the standard unit of extension.**	Metre (m)	☐
12 ★	**State how the extension of an elastic object is calculated.**	Extension = stretched length − original length	☐
13 ★	**Name the energy store that increases when an elastic object is stretched or compressed.**	The elastic potential energy store	☐
14 ★	**Describe how to find the spring constant from a graph of force (y-axis) against extension (x-axis).**	It is the gradient of the straight section.	☐
15	**Describe how to find the spring constant from a graph of extension (y-axis) against force (x-axis).**	It is 1 ÷ gradient of the straight section.	☐
16	**State the standard unit of elastic potential energy.**	Joule (J)	☐
17	**State the equation linking force, spring constant and extension.**	force = spring constant × extension ($F = k\,e$)	☐
18	**Explain why more than one force is needed to stretch or compress a stationary object.**	A single force would just make the object move.	☐
19	**State how the work done stretching a spring compares with the elastic potential energy stored.**	They are equal (if the spring is not inelastically deformed).	☐
20	**State how linear and non-linear relationships look on a force–extension graph.**	Linear: a straight line. Non-linear: a curve.	☐

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